

Algorithmic Pseudocode Sample 32

Source-backed Image2Struct algorithm sample

1 Algorithm

This sample contains algorithmic pseudocode extracted from a source-backed LaTeX benchmark dataset. It is wrapped in a minimal article document for pdfLaTeX validation.

Algorithm 1 Source-backed algorithmic procedure

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1: procedure PENALIZEDG( $\vec{A}, \vec{H}, \vec{Y}, \lambda, \text{corstr}, \kappa$ )
2:   Compute  $E(\vec{A}_i | \vec{H}_i)$  for  $i = 1, \dots, n$  ▷ logistic regression on the pooled data
3:    $\vec{\theta}^{\text{up}} \leftarrow \{\sum_{i=1}^n \vec{D}_i^\top (\vec{H}_i - \vec{A}_i \cdot \vec{H}_i)\}^{-1} \sum_{i=1}^n \vec{D}_i^\top \vec{Y}_i$  ▷ univariate unpenalized G-estimator
4:   for all  $\lambda \in (\lambda_{\max}, \dots, \lambda_{\min})$  do
5:     Initialize:  $t = 0, \vec{\theta}^0 \leftarrow \vec{\theta}^{\text{up}}$ 
6:     repeat
7:        $\vec{e}_i \leftarrow \vec{Y}_i - (\vec{H}_i - \vec{A}_i \cdot \vec{H}_i) \vec{\theta}^t$  for  $i = 1, \dots, n$ 
8:       Compute  $\sigma^t$  and  $\alpha^t$  under corstr using ?? ▷ method of moments estimator
9:       Compute  $\hat{V}_i$  according to corstr for  $i = 1, \dots, n$  ▷  $\hat{V}_i = \hat{V} \forall i$  if  $J$  is fixed
10:      Compute  $\vec{S}^{\text{eff}}(\vec{\theta}^t)$  using (??)
11:      Compute  $\vec{H}_n(\vec{\theta}^t)$  and  $\vec{E}_n(\vec{\theta}^t)$  using (??) and (??), respectively.
12:      Update  $\vec{\theta}^t$  according to (??) and obtain  $\vec{\theta}^{t+1}$ 
13:       $t \leftarrow t + 1$ 
14:    until  $\|\vec{\theta}^t - \vec{\theta}^{t-1}\| < \kappa$  ▷ we set  $\kappa = 10^{-4}$ 
15:     $\tilde{\theta}_\lambda \leftarrow \vec{\theta}^t, \tilde{\sigma}_\lambda \leftarrow \sigma^t$ , and  $\tilde{\alpha}_\lambda \leftarrow \alpha^t$ 
16:    Compute  $\text{DRIC}_\lambda$  according to (??)
17:  end for
18:   $\tilde{\theta} \leftarrow \tilde{\theta}_\lambda^*$  such that  $\tilde{\theta}_\lambda^*$  corresponds to the minimum of  $\text{DRIC}_\lambda$ 
19:  return  $\tilde{\theta}$ 
20: end procedure

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